

C++ Module TS Issues List

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Active Issues

[\[5\] Static local variables \[John Spicer, 11/8/2016\]](#)

Question

Should there be a restriction on local statics in exported function template?

Proposed Resolution

The current design purposefully does **not** place any restriction. It even has an explicit note to that effect for inline functions. Maybe that note should be clarified that it also applies to function templates.

[\[7\] Default arguments for exported declarations \[John Spicer, 11/8/2016\]](#)

Question

Consider

```
// interface unit of M
module M;
export namespace N {
    int f(int);
}
namespace N {
    int f(int = 5);
}

// 1.cxx, not part of M
import M;
int main() { return N::f(); } // OK?
```

Proposed Resolution

Only default arguments in exported declarations are effectively exported, i.e. visible to importing translation units. This issue is resolved by the new wording for issue #4. Add a note to the specification.

[\[9\] Point of definition of implicitly defined special member functions \[Roger Orr, 11/7/2016\]](#)

Question

What is the point of definition of delayed implicitly defined special member functions?

Proposed Resolution

When the definition of an implicitly defined special member function is needed, the context of the definition shall be right after the last exported declaration from the owning module's interface unit.

[10] Annotation of module declaration in module interface unit [Nathan Sidwell, 2/2/2017]

See discussion on the ‘modules’ reflector with title ‘modules’. Request:

It would be nice if the module declaration was decorated in some unique way to denote the interface unit.

Proposed Resolution

From specification perspective, there is no ambiguity about which translation unit is a module interface unit. It might however be convenient for some to see a redundant annotation in the source code indicating that a source file is indeed a module interface unit. See proposal [P0273R1](#), proposal P0584R0 titled “Module Interface and Preamble”, and proposal [P0629](#).

[11] Redeclaration within the purview of a module [Nathan Sidwell 1/9/2017]

See discussion titled ‘modules’ on the ‘modules’ reflector.

Question

In the purview of a module, can a redeclaration export an entity that wasn’t previously declared as exported?

Proposed Resolution

No. This was expressly forbidden, and that restriction is encoded via the forbidden change of linkage (from module linkage to external linkage).

Add a note in the specification to illustrate this.

[12] Exported partial specialization [Nathan Sidwell, 1/30/2017]

See the discussion ‘export and templates’ on the ‘modules’ reflector.

Question

Given an exported class template declaration X, is it possible to declare two non-exported partial specialization of X in two modules?

Proposed Resolution

Yes; however, the partial specialization shall depend on non-exported types or templates.

[13] Module Partitions [EWG, 3/4/2017]

This issue results from a EWG straw poll at the Spring 2017 Kona meeting.

Question

Provide a notation for expressing “module partitions”, see proposal [P0273R1](#) for background. See proposal P0584R0.

[14] Uniqueness of namespace names across modules [Nathan Sidwell, 4/14/2017]

Report

From the TS I see that namespaces are always exported. But can an unrelated module bind the same name to something other than a namespace? For instance

```

/// file A
module A [[interface]];
namespace N { export void Foo (); };

```

```

// file B
module B [[interface]];
export void N ();

```

```

// file C
module C [[interface]];
import A;
export inline void Frob () {N::Foo ();}

```

```

// file D
import C;
import B;

```

Is this well formed? If it isn't at what point is a diagnostic required? This case looks like it might be an ODR violation because both 'namespace N' and 'void N ()' are exported. But what if B didn't export 'void N ()'?

More philosophically, I'd conceptualized namespaces as a program-wide object, of which a particular module might have an empty partition. But, I'm not sure that's justifiable from the TS. An empty partition is not necessarily the same as a null partition. Does the latter permits a module to bind the identifier to something else?

Discussion

[\[15\] Semantics of exported using-directives \[Nathan Sidwell, 6/19/2017\]](#)

Question

What is the semantics of exported using-directive?

[\[16\] Semantics of exported using-declarations \[Nathan Sidwell, 6/19/2017\]](#)

Question

What is the semantics of exported using-declaration?

[\[17\] export of non-exported entity \[Richard Smith, 10/6/2017\]](#)

[\[18\] Investigate TS Ballot Comment US 004 \[11/08/2017\]](#)

Alisdair Meredith to provide drafting.

[19] Investigate TS Ballot Comment GB 016 [11/08/2017]

As described in TS Ballot Comment GB 016.

[20] Investigate TS Ballot Comment US 042 [11/08/2017]

As described in TS Ballot Comment US 042.

[21] Investigate TS Ballot Comment US 078 [11/08/2017]

As described in TS Ballot Comment US 078.

[22] Investigate TS Ballot Comment US 107 [11/08/2017]

As described in TS Ballot Comment US 107.

[23] Investigate TS Ballot Comment US 108 [11/08/2017]

As described in TS Ballot Comment US 108.

[24] Investigate TS Ballot Comment CA 124 [11/08/2017]

As described in TS Ballot Comment CA 124.

[25] Translation Units is not correctly used in the TS+IS document [11/09/2017]

[26] Investigate uses of 'visible' when 'reachable' is intended [11/09/2017]

[27] Check for uses of 'prior' in the base standard document [11/09/2017]

[28] some non-exported name may have module linkage [Richard Smith, 11/09/2017]

6.5p4 makes non-exported name have module linkage

[29] Internal-linkage functions and ADL [Nathan Sidwell, 1/25/2018]

Should internal-linkage functions declared in the purview of a module interface unit be visible to instantiation-time ADL of dependent call expressions? Does the answer depend on whether the instantiation context is the module-interface purview or elsewhere?

Example:

```
// TU Foo
export module Foo;
template <typename T>
void Quux (T& r) {
    Frob (r);           // #1
}

// TU Bar
export module Bar;
import Foo;
export class X { ... };
static void Frob (int) {...}; // #2
```

```

void Baz (X& x) {
    Quux (x); // #3
}

// TU Toto
import Foo;
import Bar;
void Zorch (X& x) {
    Quux (x); // #4
}

```

- a) Is #2 found during instantiation Quux<X> at #4 by ADL of the call #1
- b) Is #2 found during instantiation Quux<X> at #3 by ADL of the call #1
- c) Does the #b answer change if the declaration of Frob is after the point of instantiation #3? (i.e. if definition instantiation can be deferred to the end of compilation)

[Assume #a and #b are in different programs so we don't get into the issue of the instantiation contexts (potentially) seeing different things.]

[\[30\] Reachability of nested typedef \[1/29/2018\]](#)

Harmonize wording and example 32 on the 2018-01-29 wiki page.

[\[31\] Investigate more uses of abstract semantics graph \[Hubert Tong, 1/29/2018\]](#)

Investigate more uses of abstract semantics graph. See also issue #31.

Davis Herring: Wording is missing that says types are incomplete by default except those reachable through a certain set of rules.

[\[32\] Proclaim ownership: Amend example in 10.7.5/2 \[Alisdair Meredith, 1/30/2018\]](#)

Amend example in paragraph 10.7.5/2 to indicate how to write proclaimed ownership at namespace scope, as well as restrictions.

Closed Issues

[\[1\] export import M; \[Richard Smith, 9/7/2016\]](#)

Remove from grammar. Or ban it through semantics prose. Same thing with export { import M; }

Resolution

See [P0500R0](#) adopted at the Fall 2016 Issaquah meeting.

[\[2\] import M; at interface level \[Richard Smith, 9/7/2016\]](#)

Ban it from interface units.

Resolution

See [P0500R0](#) adopted at the Fall 2016 Issaquah meeting.

[3] `export const int n = 5;` [Richard Smith, 9/8/2016]

Clarify that this is allowed.

Resolution

See [P0500R0](#) adopted at the Fall 2016 Issaquah meeting.

[4] Import declaration and namespace partitions; [Lukasz Mendakiewicz, 11/3/2016]

Problem:

I was reading N4610 and have a question:

```
module M;
export namespace N
{
    struct A {};
}
namespace N
{
    struct B {};
}
```

7.7.1/4 says that all members of **namespace-body** are exported, meaning `N::A`.

```
import M;
```

7.7.2/1 says that import declaration adds the **namespace partitions** with external linkage from M to the current TU.

Namespace partition N from M contains both `N::A` and `N::B`.

So is `N::B` visible and can be used in the second TU or not?

Resolution

See [P0500R0](#) adopted at the Fall 2016 Issaquah meeting.

[6] Entities referenced from exported templates [John Spicer, 11/8/2016]

Question

Are there any restrictions on the linkage of the entities that can be referenced from an exported template definition?

Resolution

For function templates, by design, **there is no restriction on the linkage of entities that can be referenced from a definition**. Note, this is the same restriction as for inline functions.

[8] Point of instantiation across translation units and name lookup [John Spicer, 11/8/2016]

See paper P0582R0 titled "*Modules: Contexts of Template Instantiations and Name Lookup*", adopted at the Spring 2017 Kona meeting.